VAD guided noise updating for Wiener speech enhancement

NOIZEUS benchmarks and controlled changes in noise level

Yulong Chen | SID 550889472 | GitHub yche0441
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Project website and demonstration materials

Repository and downloadable code

Video presentation and code demonstration

Abstract

This project tests when voice activity detection (VAD) helps a Wiener speech enhancer update its noise estimate. A shared two-step noise reduction (TSNR) path compares fixed estimation with an energy gate, then tests a more protective gate selected on development data. All main intelligibility results use one standard STOI implementation. In 32 original evaluation conditions, adaptive updating improves both reference SNR and STOI in only five cases. A nine-setting development study selects a 1 dB threshold and ten-frame hangover. On 48 unused sentence conditions, this setting adds 0.00404 mean STOI over the original gate but remains 0.00064 below fixed estimation. The direction of each average survives removing any one test sentence. A further controlled experiment reuses six sentences and identical underlying noise, applying constant, rising and falling noise envelopes without retuning. It directly examines how gate decisions affect tracking after a level change. Neither gate permits late updates in any rising-noise case, whereas falling noise allows renewed tracking; the selected method then gains 0.143 dB and 0.00201 STOI over fixed estimation. The evidence supports conditional benefits from adaptation, with a trade-off between protecting speech and allowing noise updates. The measurements establish neither perceptual preference nor general superiority.

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1 Research question and contribution

Problem and motivation

Single-channel speech enhancement must estimate speech and noise from their mixture. Reducing background energy can also remove weak speech. A useful investigation therefore needs a reference signal, complementary metrics and an explanation of the conditions in which the processing helps or fails.

A Wiener enhancer uses a noise power spectral density (PSD) estimate in every frequency bin. Holding the initial estimate fixed is stable but cannot track later changes. Updating during detected pauses can track noise if suitable frames are admitted. An overestimated noise PSD tends to lower the estimated speech-to-noise ratio and gain, risking weak-speech suppression; an underestimate can leave more residual noise. Weak speech admitted as noise and loud noise mistaken for speech create different estimation failures.

Research question

When does energy-VAD-guided noise updating improve a TSNR Wiener enhancer compared with a fixed initial estimate, and how do speech protection and noise tracking affect that result? The scope is one established enhancement path and one inspectable update rule. The study examines the rule rather than ranking complete speech enhancement systems.

Contribution

The TSNR core derives from LIU Ming and Pascal Scalart [6] and the method of Plapous et al. [1]. The project contribution is a reproducible investigation of its noise-update policy. A common amplitude-preserving reconstruction makes the comparisons suitable for reference scoring. Section 6 selects a more protective gate using development data, then tests it on unused sentences. Sections 8 and 9 hold the speech and underlying noise fixed while imposing level changes, linking score differences to frame-level update and tracking behaviour. These experiments identify when this particular update rule helps or fails; they make no claim to a new TSNR algorithm.

The analysis uses sample-aligned clean and noisy NOIZEUS recordings [2,5]. All methods receive the same noisy signal and use the same window, initialisation, gain calculation and reconstruction. The clean signal is available only to scoring and diagnostic routines. The original data, source licence and numerical outputs remain available in the accompanying code package.

What the experiments establish

The original gate makes small average SNR gains but loses average STOI against fixed estimation. Lowering the decision threshold and lengthening the hangover protects more speech-associated frames and improves mean STOI relative to that gate. The selected rule still updates infrequently and does not exceed fixed estimation on average STOI. Controlled changes in noise level test whether this limited updating also restricts tracking. These are conditional findings about the stated algorithm, recordings and imposed changes.

The report distinguishes the original evaluation, the development-selected new-sentence test, and the later diagnostic experiment. Reused sentences and retrospective analyses are identified explicitly. Each stage answers a different question; additional noise conditions do not create additional independent speakers.

2 Literature and design rationale

The enhancement path

Scalart and Vieira Filho [7] make a priori SNR estimation central to speech enhancement: errors in estimated speech or noise power affect the Wiener gain. Plapous et al. [1] use TSNR to refine the delayed decision-directed estimate. Their harmonic-regeneration extension is not implemented here. Keeping TSNR fixed isolates the effect of the noise-update policy.

Choosing a noise estimator

Sohn et al. [8] use likelihood-ratio decisions and temporal modelling for statistical VAD. The present broadband energy gate uses less structure, but its threshold and hangover expose the trade-off between protecting speech and admitting updates. It does not reproduce the Sohn detector.

Martin [9] tracks smoothed spectral minima; MCRA [4] and improved MCRA [10] regulate averaging using speech-presence information. Gerkmann and Hendriks [12] use a soft speech-presence probability with fixed priors, separating noise estimation from subsequent speech-power estimation. These approaches retain spectral history or uncertainty that a binary energy decision discards. They motivate a targeted follow-up: replace only the noise tracker within the shared TSNR path and test tracking and speech distortion together. Their performance is not measured here.

Estimator	Useful property	Principal limitation here
Fixed initial PSD	Stable common baseline	Cannot follow later changes
Energy VAD updating	Simple decisions and explicit update times	Speech contamination or blocked tracking
Minima or probability based tracking [4,9,10,12]	Uses spectral history or graded speech evidence	Additional modelling choices; untested here

The hard gate provides an inspectable first experiment: its decisions reveal when adaptation is allowed, and the matched noise-level changes test the resulting tracking behaviour. This keeps the research question focused while establishing a reason to investigate a more elaborate estimator.

Relation to recent enhancement systems

DeepFilterNet [13] combines learned ERB-band gains for the spectral envelope with complex multi-frame filtering for speech periodicity, demonstrated on 48 kHz audio. Its learned representation and reconstruction differ from the present 8 kHz TSNR pipeline. A direct comparison would assess complete systems rather than isolate noise updating. The present study therefore provides an interpretable component investigation; it does not establish a ranking against learned models or claims about their robustness.

What the measurements mean

Hu and Loizou [2] distinguish speech distortion, background-noise distortion and overall quality. Taal et al. [3] introduce STOI as an intelligibility predictor for noisy and time-frequency processed speech. Reference SNR and STOI therefore provide complementary evidence; STOI is neither a word-recognition percentage nor a listening-quality score.

Two distinctions guide the experiment: the noise PSD describes the estimated background, whereas a priori SNR controls attenuation using that estimate; a lower waveform error need not imply higher intelligibility. Paired metrics and update traces test these different aspects. References [5,6,11] identify the data and software used.

3 Processing method

Common TSNR analysis and synthesis

For additive, uncorrelated speech and noise, the minimum-mean-square-error linear spectral gain is speech power divided by total power, equivalently ξ / (1 + ξ) for a priori speech-to-noise ratio ξ [7]. The implemented gains use estimated powers and additional constraints [1], so the ideal model does not guarantee minimum error on each recording.

The input is mono at 8 kHz. A 200-sample window, 80-sample hop and 400-point FFT implement 25 ms analysis and 10 ms updates. The 20 Hz FFT-bin spacing includes zero padding; the 25 ms window and its shape determine frequency resolving power. The source Hann convention is w[n] = 0.5 - 0.5 cos(2 pi n / 201), n = 1,...,200. All methods share these settings.

Let $Y_t[k]$ denote a noisy-frame spectrum and $D_t[k]$ the current noise PSD. The source branch forms a floored instantaneous estimate $q_t = \max(|Y_t|^2 / D_t - 1, 0.1)$. The decision-directed estimate combines q_t with the previous enhanced magnitude using a 0.99 weight on the previous estimate. The following expressions apply separately in each frequency bin:

$$\xi_t = 0.99\ |\hat{S}_{t-1}|^2\ / D_t + 0.01\, q_t$$

$$G_{1,t} = \xi_t / \left(1 + \xi_t\right); \xi_{2,t} = |G_{1,t} Y_t|^2\ / D_t$$

A second Wiener gain $\xi_{2,t} / (1 + \xi_{2,t})$ has a pre-constraint floor of 0.15. The original impulse-response constraint and energy normalisation are then applied, so 0.15 is not asserted to bound every final constrained gain. The observed phase is retained. Both modes use identical inverse-FFT accumulation, window-sum normalisation, padding and cropping.

Noise initialisation and updating

Every mode starts from the same mean power spectrum of 601 windows advanced one sample at a time within the first 0.10 s. Fixed estimation retains this PSD. The gate computes a broadband energy ratio, with each sum taken over the FFT bins:

$$R_t = 10 \log_{10}\left(\Sigma\ |Y_t[k]|^2\ /\ \Sigma D_t[k]\right)$$

A ratio R_t greater than threshold θ declares speech and resets the hangover to H frames. The next H frames at or below threshold remain protected; another detection restarts the count. Protected frames retain D_t. Eligible unprotected frames update only after current-frame enhancement:

$$D_{t+1} = 0.95\, D_t + 0.05\ |Y_t|^2$$

The original gate uses 3 dB and five hangover frames. The selected gate uses 1 dB and ten frames. A lower threshold declares speech more readily; a longer hangover retains protection after a detection. Calibration and padded frames never update. Their clean reference is never an algorithm input.

Reconstruction and attribution

Independent peak normalisation from the source demo is removed because it would alter reference-error comparisons. With no additional synthesis window, overlap-add is divided by the shifted analysis-window sum, not its squared sum. The unity-gain check validates amplitude and alignment, not enhancement quality. A separate legacy mode reproduces the source for verification. The licence and source headers credit the inherited TSNR implementation [6]; the common reconstruction and update experiments are documented adaptations.

4 Experimental design and verification

Three complementary stages

Stage	Material and selection	Question
Original benchmark	sp01/sp16 development; sp06/sp11/sp21/sp26 evaluation; car/street at 0/5/10/15 dB	Does the original gate help fixed TSNR?
Parameter study	9 settings on 16 development cases; selected rule frozen before 48 cases from six unused sentences	Does a more protective gate improve the original rule?
Controlled diagnostic	Same six test sentences reused; same residual noise under 3 prescribed level profiles; 36 mixtures	How do imposed changes interact with the frozen gates?

The original 32 evaluation conditions span four other speakers than the two development sentences. The later test uses sp02, sp07, sp12, sp17, sp22 and sp27: unused sentences at selection time, but speakers already represented in the corpus subsets. The controlled diagnostic deliberately reuses these sentences. Its protocol is saved before running, with no further parameter selection.

One intelligibility implementation throughout

Every main STOI value in this report uses standard pystoi 0.4.1, extended=False [11]. The original v1 outputs have been retrospectively rescored from unchanged, unquantized waveforms. Historical native MATLAB scores remain in the archive. This rescore changes one paired STOI direction and the count improving both metrics from six to five of 32; neither algorithm nor its output is changed.

Across 144 archived scores, mean absolute implementation disagreement is 0.00285; maximum paired-effect disagreement is 0.00542. This motivates consistent scoring, not an error bar on within-pystoi comparisons. The new study and diagnostic already use the common implementation.

Scoring and interpretation

Reference SNR is 10 log10 of clean energy divided by processed-minus-clean error energy. It penalises residual noise and speech distortion together. All whole-recording metrics exclude samples 1:800; there is no delay fitting, gain fitting or per-method amplitude normalisation. Nominal NOIZEUS SNR labels use active speech scaling [5] and need not equal this measured SNR. Means are arithmetic means of case scores, not pooled waveform SNR.

Paired differences always use the same input and reference. New-test sensitivity averages the eight conditions within each sentence, then removes one entire sentence at a time. These six leave-one-out averages are descriptive checks, not confidence intervals or significance tests. The controlled experiment likewise reports all six sentence summaries rather than treating its 36 conditions as independent speakers.

Diagnostic definitions

Valid frames start after sample 800 and contain no analysis padding. Relative to each recording's peak valid clean-frame energy, frames at or above -30 dB form the speech proxy; those at or below -40 dB form the low-energy proxy. Overlap pools updated speech-proxy frames and divides by all speech-proxy frames across cases. Coverage instead averages each case's updated/valid-frame fraction. Thus 16% overlap does not mean 16% of updates contain speech. These labels are diagnostic proxies, never detector inputs.

Verification

Checks cover source reproduction, unity reconstruction, identical initial PSDs, frozen-update equivalence, finite metrics and result-table arithmetic. The original reference SNR was independently reproduced within 6×10^{-14} dB. New diagnostic checks additionally require identical pre-change mixtures, PSDs and traces across paired profiles. Enhancement uses MATLAB R2026a; standard STOI uses Python on unquantized arrays. Saved code, protocols, checks and per-case results support reproduction.

5 Original benchmark reassessed

Table 1 uses the four evaluation sentences, with 16 cases per noise and method. The SNRs are unchanged; all STOI values use the common retrospective rescore. Both enhanced methods raise average SNR over the input, but the incremental gain from updating is small.

Table 1 Original evaluation means with standard STOI

Noise	Method	SNR dB	STOI
Car	Input	7.103	0.7968
Car	Fixed	10.814	0.7966
Car	Original VAD	10.863	0.7894
Street	Input	7.207	0.8236
Street	Fixed	9.656	0.8034
Street	Original VAD	9.800	0.8023

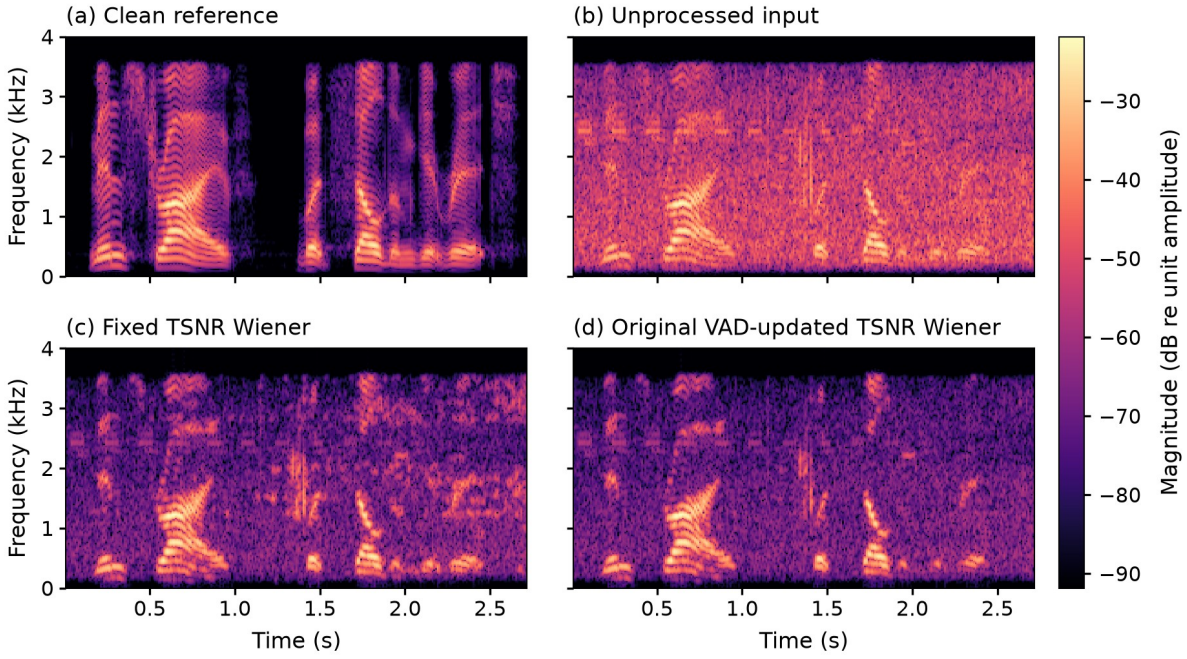


Figure 1 sp06 in street noise at nominal 5 dB. The clean, input, fixed and original VAD panels share the analysis settings and 70 dB colour range. These spectra are unchanged by the metric rescore.

For this example, original VAD raises SNR from 9.793 to 9.897 dB relative to fixed estimation, but STOI falls from 0.7539 to 0.7424. The darker background accompanies lower waveform error without higher predicted intelligibility. In the corresponding trace, updates overlap 42 of 166 clean speech-energy proxy frames; this is a diagnostic clue, not proof of the cause of every score change.

Across the evaluation set, original VAD changes mean SNR by +0.049 dB in car and +0.144 dB in street noise. Mean STOI changes by -0.00717 and -0.00103. It improves SNR in 18/32 cases, STOI in 8/32, and both in 5/32. The small mean SNR advantage therefore cannot establish a generally better enhancer.

6 Parameter selection and new sentence test

The extension tests a more protective gate while retaining the 0.95 PSD smoother. Thresholds 1, 3 and 5 dB cross hangovers of 0, 5 and 10 frames on the 16 sp01/sp16 development cases. The selection rule maximises mean development STOI; ties within 10^{-6} use higher SNR, then proximity to the original settings. The rule and test IDs were recorded before scoring.

Table 2 Development results and pooled speech-proxy update overlap

Threshold dB	Hangover frames	SNR dB	STOI	Overlap
1	0	11.467	0.7999	14.7%
1	5	11.375	0.8009	3.7%
1	10	11.300	0.8012	2.6%
3	0	11.256	0.7914	35.6%
3	5	11.265	0.7949	19.8%
3	10	11.227	0.7967	15.5%
5	0	10.670	0.7836	52.6%
5	5	10.748	0.7858	38.6%
5	10	10.566	0.7888	33.4%

At five-frame hangover, lowering the threshold from 3 to 1 dB adds 0.00598 STOI. At 3 dB, increasing hangover from five to ten frames adds 0.00181. At 1 dB, this extension adds only 0.00025 while reducing update coverage from 14.1% to 8.5%. Lowering the threshold provides the larger development gain in these comparisons; extra hangover supplies a smaller gain at the cost of tracking opportunities.

The rule selects 1 dB and ten frames, with development STOI 0.8012. The highest SNR instead occurs at 1 dB and zero hangover. The chosen objective therefore matters. No retuning follows the new-sentence test.

Table 3 Means over 48 new sentence conditions

Method	SNR dB	STOI	Proxy overlap
Input	7.197	0.8102	-
Fixed	10.651	0.8142	-
Original VAD	10.656	0.8095	16.0%
Selected VAD	10.715	0.8136	2.7%

Selected VAD adds 0.059 dB SNR and 0.00404 STOI over the original gate, with 33/48 STOI improvements and 19/48 improvements in both metrics. Proxy overlap falls from 16.0% to 2.7%, while update coverage falls from 27.4% to 4.9%. Against fixed estimation it adds 0.064 dB but loses 0.00064 STOI, and only 14/48 cases improve both. The modification improves the original rule without establishing superiority over fixed estimation.

7 Condition effects and sentence sensitivity

Table 4 separates selected-minus-fixed effects across the eight noise/SNR groups, each containing the same six sentences. A positive group average need not imply consistent improvement for individual sentences.

Table 4 Selected VAD minus fixed estimation

Noise	Nominal SNR dB	SNR change dB	STOI change	Both improve
Car	0	+0.0675	-0.00122	1/6
Car	5	+0.0181	-0.00015	3/6
Car	10	+0.0165	+0.00029	2/6
Car	15	+0.0358	+0.00064	4/6
Street	0	-0.0849	-0.00003	1/6
Street	5	+0.3316	-0.00596	0/6
Street	10	+0.1022	+0.00207	3/6
Street	15	+0.0219	-0.00079	0/6

The largest mean SNR gain, street at 5 dB, coincides with a STOI loss and no sentence improving both scores. Street at 10 dB improves both averages but only three of six sentences improve both. Different source excerpts can occur at different nominal SNRs, so these rows do not isolate the effect of SNR alone.

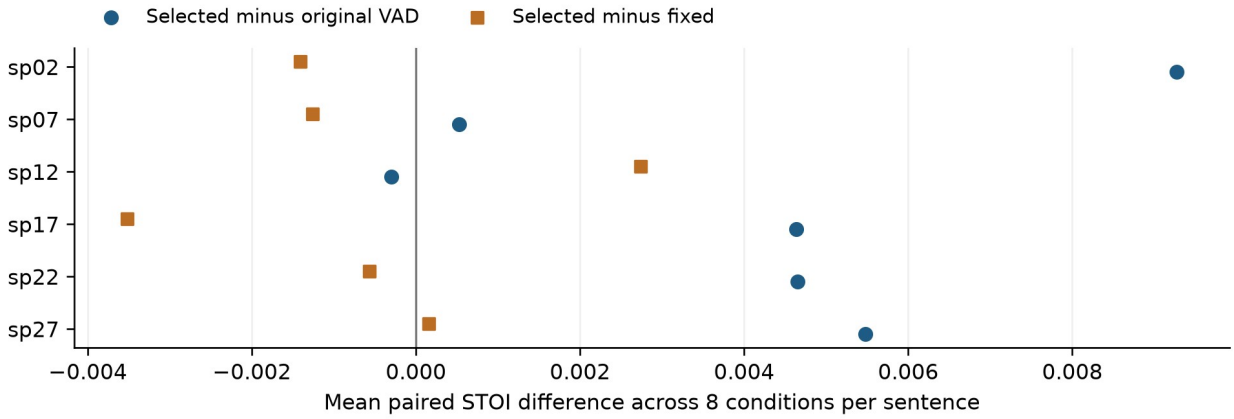


Figure 2 Every sentence-level mean paired STOI effect. Each point averages eight conditions. The six sentences, rather than the 48 repeated conditions, are the units of the sensitivity check.

Five of six sentence means favour selected over original VAD. Removing any one sentence leaves a positive mean STOI effect of +0.00300 to +0.00491. Only two sentence means favour selected over fixed; all leave-one-out averages remain negative (-0.00132 to -0.00007). The direction is not driven by one sentence, although these checks do not quantify population uncertainty.

The predetermined sp02/street/5 dB demonstration remains a counterexample: selected VAD reduces proxy overlap from 25/193 to 10/193 frames versus original VAD, yet STOI falls from 0.7278 to 0.7165. Lower proxy overlap alone therefore does not guarantee a better intelligibility score.

8 Controlled changes in noise level

The earlier noise/SNR groups use different excerpts. This diagnostic instead keeps each underlying speech and noise pair identical, then changes only the noise envelope. It reuses six test sentences and car/street mixtures at nominal 10 dB. Subtracting aligned clean speech s[t] from the noisy source gives residual r[t]. The new input is y[t] = s[t] + a[t]r[t], with no gain fitting. The residual includes any source quantisation discrepancy.

The first 100 ms and the entire pre-change prefix are unchanged. Halfway through the remaining samples, a 50 ms ramp linear in dB reaches 0, +6 or -6 dB noise amplitude level, then holds. The 36 mixtures retain an unprocessed reference and use the frozen fixed, original and selected methods. This tests prescribed trajectories, not variability at equal average SNR. All parameters and the sp02/street/rise example were recorded before this run.

Table 5 Selected minus fixed: six sentence means, each averaging car and street

Noise profile	Whole SNR dB	Whole STOI	Late SNR dB
Constant	+0.0594	+0.001180	+0.0552
Rise +6 dB	-0.0359	+0.000914	-0.0578
Fall -6 dB	+0.1428	+0.002011	+0.3949

Whole scores exclude calibration. Late SNR begins 100 ms after the ramp ends and uses the same clean reference; short late segments are not scored with STOI. Figure 3 shows all six sentence effects and their means, without confidence intervals.

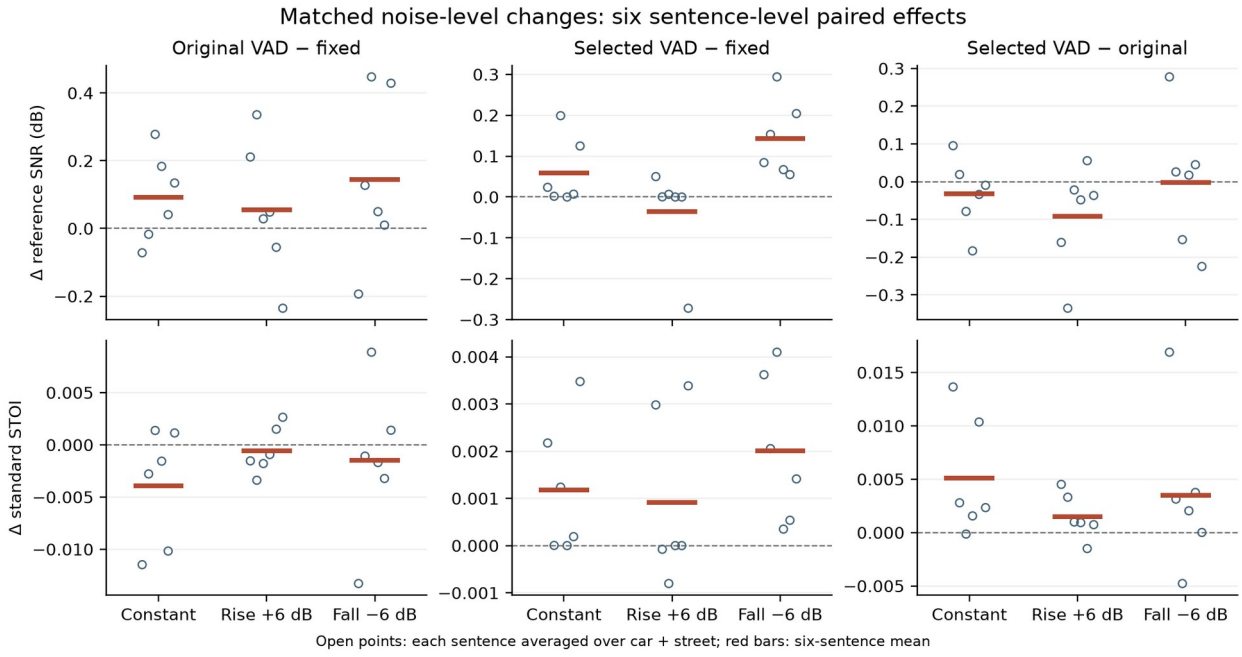


Figure 3 Matched profile comparisons. Open circles are sentence means over two noises; red bars average the six sentences. Axes differ across panels. The complete per-case results retain all outcomes.

For falling noise, selected VAD exceeds fixed SNR and STOI in all six sentence means. However, its mean STOI of 0.9059 remains below the input's 0.9071. With rising noise, selected VAD loses mean SNR against fixed despite a small STOI gain. Improvement over another processed output is therefore not a guarantee of improvement over the input.

9 Tracking behaviour and controlled results

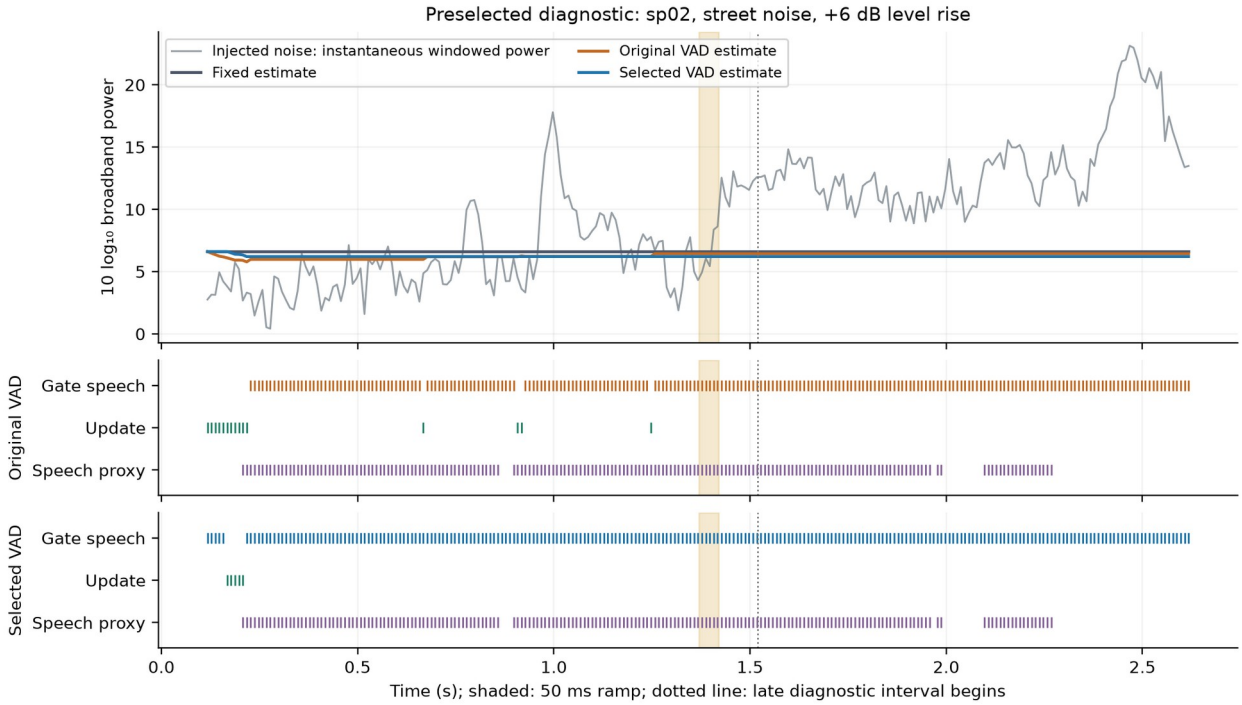


Figure 4 Prespecified sp02/street/+6 dB example. The estimator trace is read before each update. The grey line is the summed power of the known injected noise using the identical analysis window and FFT. It is a broadband realisation diagnostic, not an expected spectral PSD error.

The gate blocks the required update after the rise: both detectors declare speech throughout all 109 late frames in this example, including 27 low-clean-energy proxy frames. Across all 12 rising-noise pairs, neither VAD updates in the late interval. The mean per-case late power ratio of estimated to injected broadband power is -8.58 dB for original and -8.68 dB for selected VAD. The flat estimates in Figure 4 expose the mechanism directly.

Nonzero score differences do not demonstrate successful late tracking: the methods retain estimates and enhancement states established earlier, while whole-recording STOI also includes pre-change speech.

After a fall, mean per-case late update coverage is 34.7% for original and 17.6% for selected VAD. Mean late power ratios are +2.18 and +2.64 dB, versus +3.47 dB for fixed estimation. The lower noise allows more updates, while the protective gate still admits fewer. Each ratio compares summed estimated and injected frame powers in one late interval; these dB ratios are averaged over the 12 matched pairs. They describe the realised signal rather than expected PSD.

Even with every frame admitted, a 0.95 smoother requires approximately ln(0.1)/ln(0.95) = 45 updates, or 0.45 s at a 10 ms hop, to remove 90% of an ideal constant-target initial error. Gating extends elapsed adaptation time; short speech recordings may end before settling. This calculation explains the possible time scale, not a fitted settling-time measurement.

The 12 constant-source inputs pass identity reconstruction; all 36 mixtures pass shared-prefix and initial-PSD checks. Independent Python calculations reproduce every diagnostic noise-power frame and the before/after update recurrence within 8×10^{-15} relative error. In matched late frames, a ±6 dB injection changes the fixed-estimator log ratio by exactly ∓6 dB. The constant-profile scores reproduce the prior 10 dB test, with maximum STOI difference below 5×10^{-16}.

10 Discussion and conclusions

What helps and what it costs

The benchmark and parameter study support a consistent interpretation. A lower threshold and longer hangover protect more frames associated with clean speech. The selected gate improves average STOI over original VAD, including after removal of each test sentence, but leaves only 4.9% update coverage. It consequently behaves more like fixed estimation and retains a small mean STOI deficit against that baseline. Speech protection and adaptation are competing objectives within this rule.

The controlled diagnostic supplies a mechanism that the original noise categories could not isolate. A rise increases mixture-to-estimate energy and keeps both gates closed, leaving the noise estimate unable to follow the change. Falling noise instead permits updates, and the selected method improves SNR and STOI over fixed estimation in all six sentence means. Yet it remains below the unprocessed falling-noise mean STOI. Adaptation benefits are therefore conditional on gate access and evaluation criterion, and the effect sizes remain small.

The blocked updates provide a concrete connection to the motivation for minima-based and probability-weighted tracking [4,9,10,12]. Gerkmann and Hendriks [12] retain uncertainty through soft speech-presence probabilities; this suggests a controlled alternative to the binary gate. It remains necessary to test whether that estimator can recover from these upward changes without increasing speech distortion. Neither its published results nor those of DeepFilterNet [13] are directly comparable with these short 8 kHz cases. All present methods share TSNR, so this study also does not isolate the benefit of its second SNR refinement [1].

Limits of the evidence

The original evaluation has four sentences from four speakers; the later six sentences use known speakers, and the controlled diagnostic reuses them. Repeated conditions increase coverage without adding independent speakers. Leave-one-out averages check sensitivity, not population uncertainty. The small improvements over original VAD do not establish a noticeable perceptual benefit.

The clean-energy proxy thresholds in Section 4 are not human VAD annotations; intermediate frames remain unlabelled. Weak consonants and recording residuals complicate interpretation. Broadband power tracking describes one part of the estimator and does not establish spectral accuracy. Initial 0.10 s calibration is approximately noise-only; weak clean energy can contaminate both methods.

The imposed envelopes change later SNR and total noise energy as well as temporal variation. The results concern these trajectories, not variability at equal average SNR. Short utterances limit settling-time observations. STOI predicts intelligibility; no subjective listening evaluation or manual annotation was performed. Audio examples demonstrate outputs only. All main comparisons use the same standard STOI implementation.

Conclusion and next investigation

Energy-gated updating provides small average waveform-error improvements but does not reliably improve intelligibility predictions over fixed estimation. The development-selected gate gains 0.00404 mean STOI over original VAD on unused sentences, while remaining 0.00064 below fixed estimation. For the measured mean-STOI objective, these results do not justify replacing the fixed baseline with this gate. The matched experiment shows renewed tracking after a fall and blocked updates after a rise. The contribution is a measured explanation of the speech-protection versus tracking trade-off, supported by counterexamples to a general benefit from adaptation.

A next experiment should replace only the tracker with a probability-weighted estimator [12], retain the shared TSNR path, and use longer recordings from independently selected speakers. A separate development set would determine parameters before an untouched test. Spectral noise-estimation error, reference SNR and STOI would test tracking and distortion; human speech annotations would check the proxy labels. A blinded listening comparison would be needed before claiming perceptual benefit. These are proposed tests, not completed outcomes or guaranteed improvements.

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